

RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI-602105

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

Submitted by

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in

COMPUTER SCIENCE AND ENGINEERING

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EXPERIMENT 1

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

To study the concept of computer forensics and understand the different tools used for forensic investigation.

INTRODUCTION

Computer forensics is a branch of digital forensics that deals with identifying, collecting, preserving, and analyzing digital evidence from computers and other digital devices.

It is used to investigate cybercrimes, unauthorized access, data theft, and other security incidents. Digital evidence may include log files, deleted files, browser history, system activity, and other information stored on a computer.

THEORY

Digital forensic investigators examine computers to find evidence that helps them understand what happened during a security incident. Even when files are deleted, some information may remain and can sometimes be recovered.

The main steps in a digital forensic investigation are:

1. Identification: Identifying the devices and possible sources of digital evidence.
2. Collection: Collecting the relevant evidence carefully.
3. Preservation: Protecting the evidence from alteration or damage.
4. Examination: Examining the collected data using forensic tools.
5. Analysis: Interpreting the evidence to understand the incident.
6. Presentation: Preparing a report of the findings.

DIFFERENT TOOLS USED IN COMPUTER FORENSICS

1. EnCase Forensic

EnCase is a commercial forensic tool used to examine digital evidence, investigate computer incidents, and prepare forensic reports.

2. FTK (Forensic Toolkit)

FTK is a digital forensic tool used to analyze digital evidence, search files, and investigate information stored on computers.

3. ProDiscover

ProDiscover is a forensic investigation tool used to examine computer storage devices and analyze digital evidence.

4. Autopsy

Autopsy is a digital forensic platform that provides a graphical interface for examining digital evidence. It can help investigators analyze files, recover certain deleted files, and examine user activity.

5. The Sleuth Kit (TSK)

The Sleuth Kit is a collection of command-line tools used to examine file systems and analyze digital evidence.

6. FTK Imager

FTK Imager is used to create forensic images of storage devices and examine their contents. A forensic image is a copy of digital data created for investigation.

7. USBDeview

USBDeview is a utility that displays information about USB devices connected to a Windows computer, including previously connected devices.

PROCEDURE:

Studied the basic concepts and importance of computer forensics.

1. Identified the major stages of a digital forensic investigation.
2. Studied the functions of different forensic tools, including EnCase, FTK, ProDiscover, Autopsy, The Sleuth Kit, FTK Imager, and USBDeview.
3. Understood how these tools help investigators examine digital evidence and investigate security incidents.

RESULT

Thus, the concept of computer forensics, the stages of digital forensic investigation, and the functions of different forensic tools were studied successfully.

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EXPERIMENT 2

LIVE FORENSICS CASE INVESTIGATION USING AUTOPSY

AIM

To investigate digital evidence using Autopsy and examine files, metadata, and deleted-file information.

PROCEDURE

1. Open Autopsy and select Create New Case.

2. Enter the case details:

- Case Name: Live_Forensics
- Case Number: LF001
- Examiner Name: Your Name

3. Click Finish to create the case.

4. Select Add Data Source.

5. Choose Local Disk and select the authorized test computer's C: drive or the available forensic image.

6. Select the required ingest modules, such as Recent Activity, File Type Identification, Keyword Search and Hash Lookup.

7. Click Next or Finish to start the ingest process and wait for the analysis to complete.

8. Open Data Sources and examine the available user folders, such as Desktop, Documents, Downloads and Pictures.

9. Navigate to Results → Recent Activity to examine available browser activity and other recorded user activities.

10. Use Keyword Search to search for relevant terms, if the module is enabled and configured.

11. Open the Deleted Files section, if available, and examine the identified deleted files.

12. Select an important file and examine its details, such as file path, size, created time, modified time and accessed time, wherever available.

13. Record the relevant evidence and timestamps in the observation table.

14. Tag important evidence as an Interesting Item or Notable Item, depending on the available options.

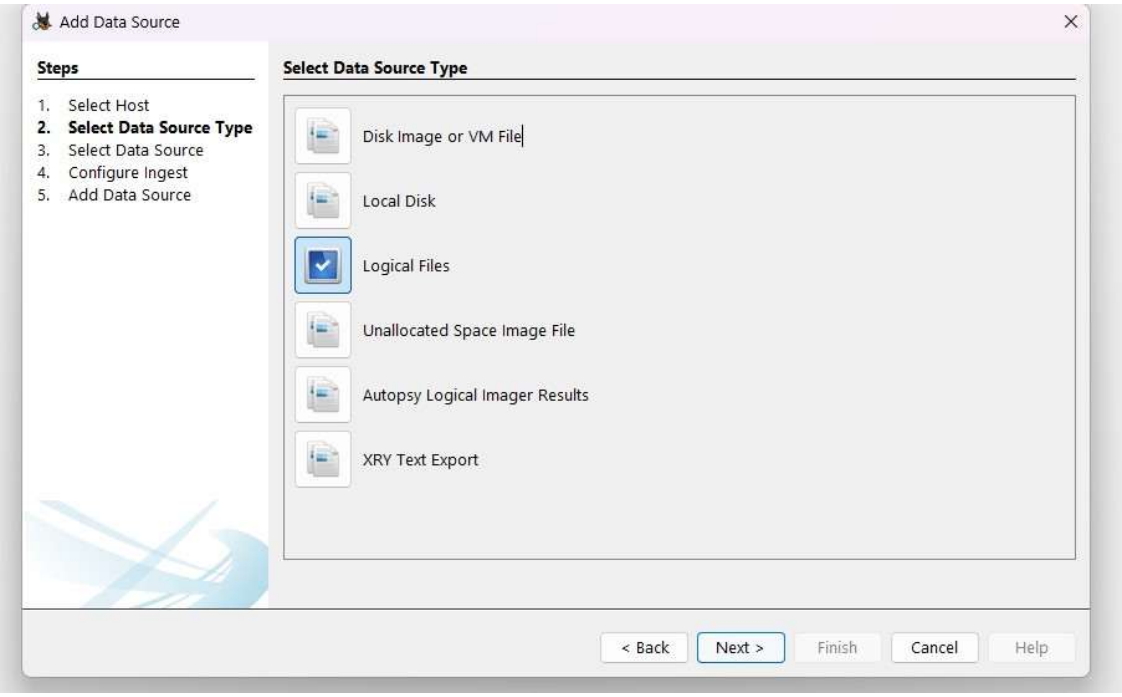
15. Generate an HTML report using Tools → Generate Report, if supported by the installed version.

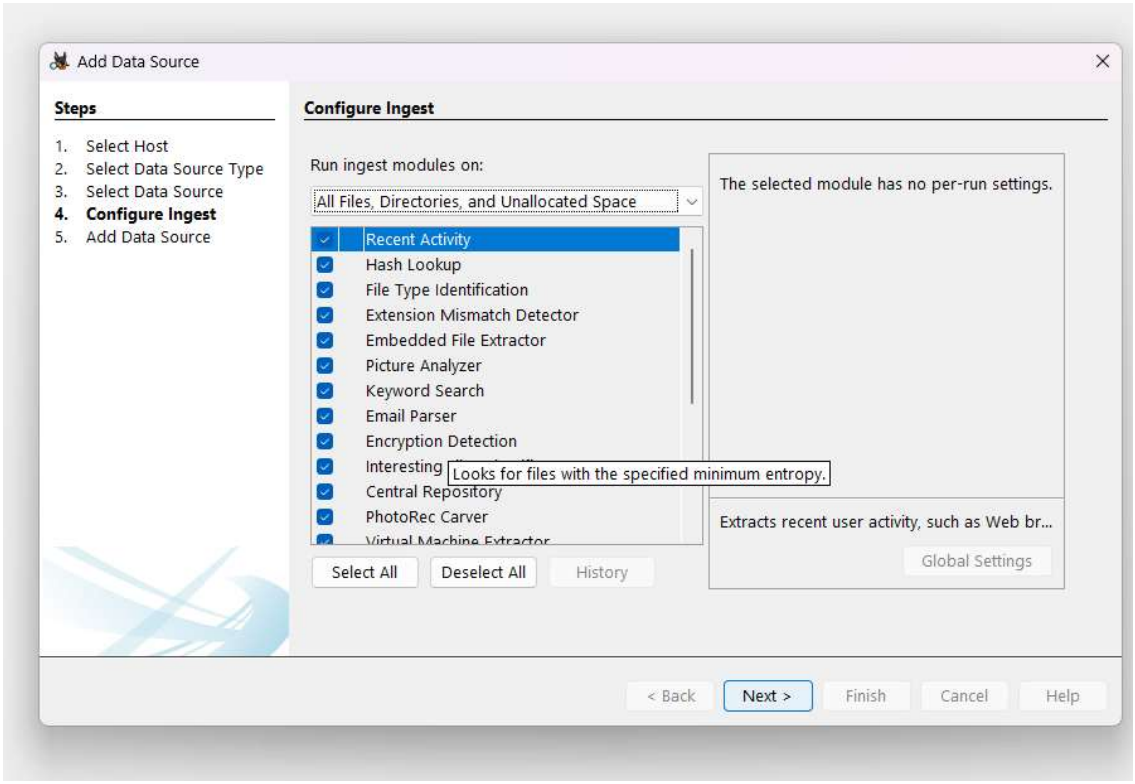
16. Open the generated report and verify the documented findings.

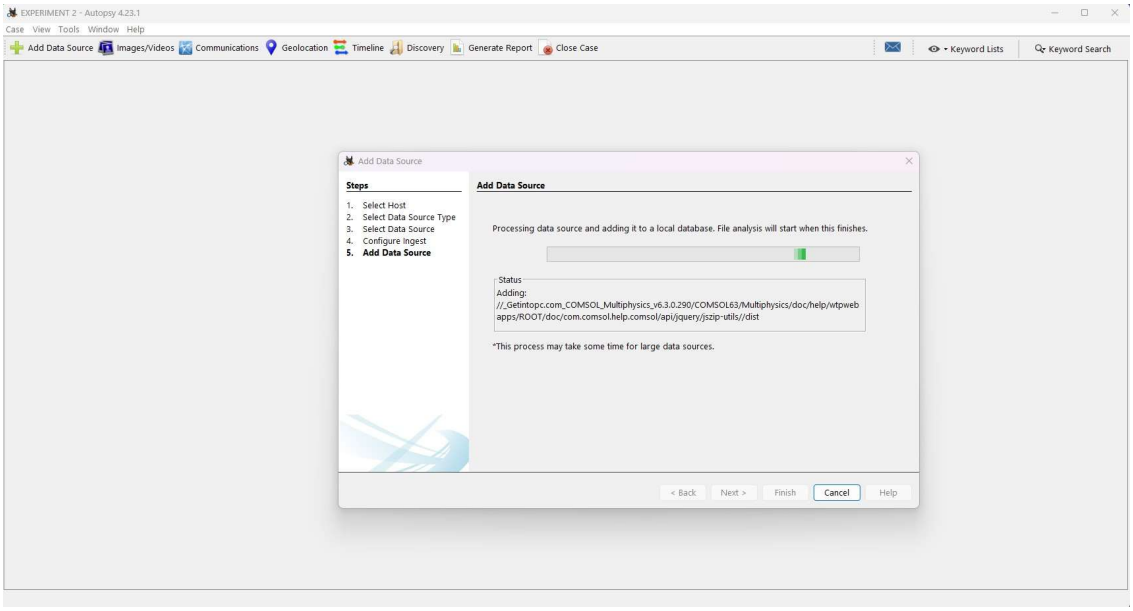
OBSERVATION

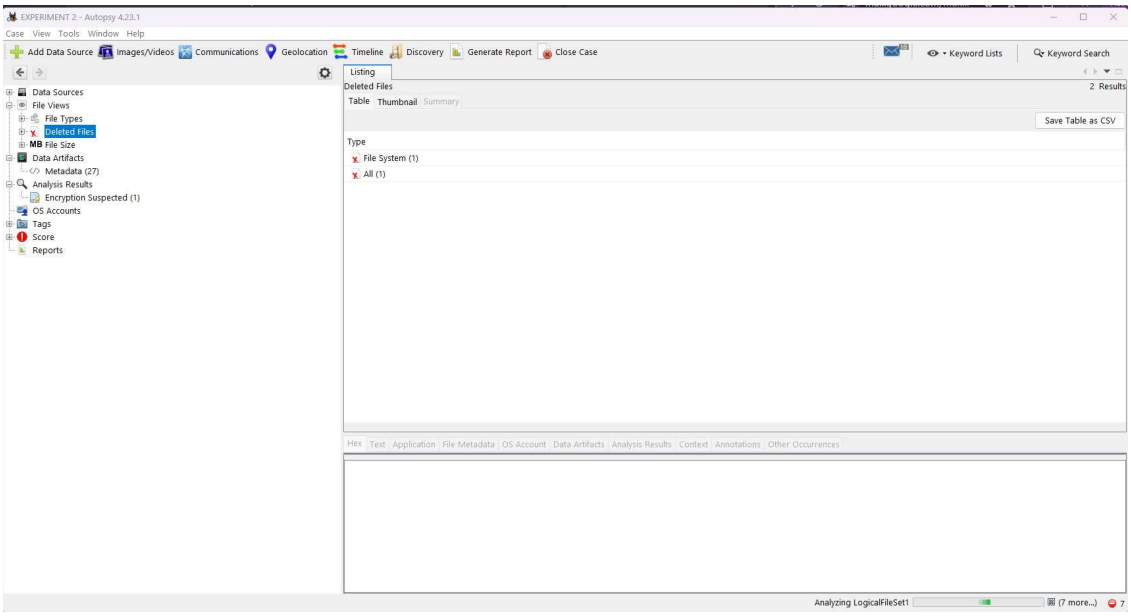
Autopsy processed the selected logical files and displayed forensic information such as file metadata, creation and modification dates, and deleted-file entries.

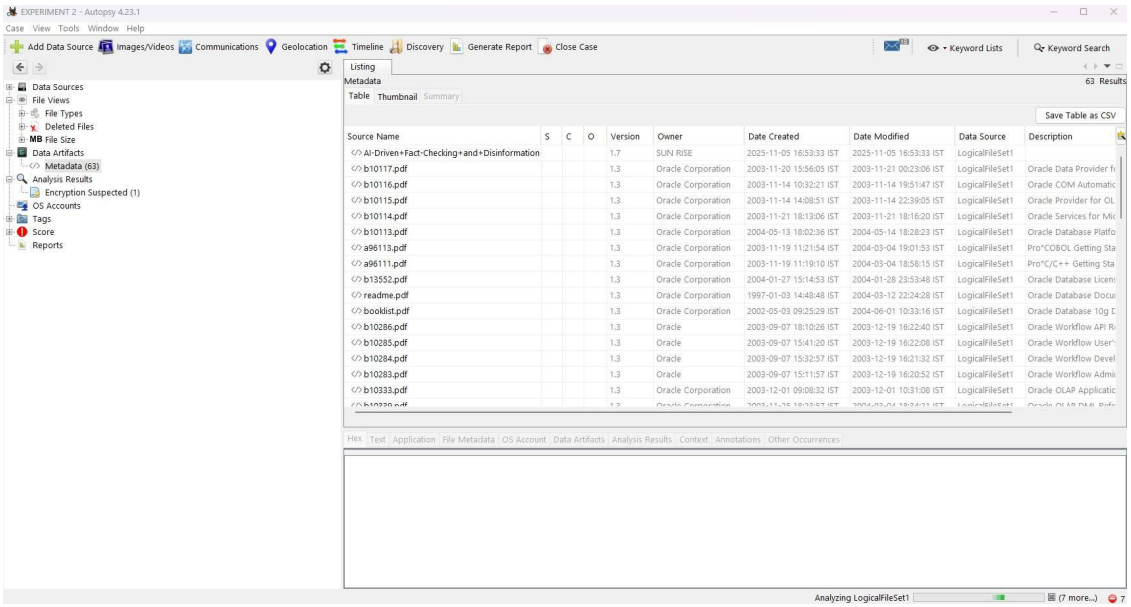
EVIDENCE EXAMINED	FINDING
User files	Files were identified in Desktop, Documents, Downloads and Pictures folders.
Browser history	Recent browsing activities were identified, where available.
Downloaded files	Downloaded files were identified and their locations were examined.
Keyword search	Keyword search results were examined for matching terms.
Deleted files	Deleted files were identified, if available.
File timestamps	File creation and modification timestamps were examined using Autopsy.
Tagged evidence	Important evidence items were tagged successfully.




Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting
Looks for files with the specified minimum entropy.
Central Repository
PhotoRec Carver
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web br...
Global Settings
< Back
Next >
Finish
Cancel
Help


Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Processing data source and adding it to a local database. File analysis will start when this finishes.
Status
Adding:
*This process may take some time for large data sources.
< Back
Next >
Finish
Cancel
Help





RESULT

Thus, digital evidence was examined using Autopsy, and the available file metadata and deleted-file information were observed.

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EXPERIMENT 3

RECOVER DELETED FILES USING FORENSICS TOOLS

AIM

To recover deleted files from a storage device using forensic tools and examine the recovered data.

PROCEDURE

1. Acquire Image of USB Drive
    - Create a forensic image of the USB drive using FTK Imager or the dd command.
    - Ensure write-blocking to maintain data integrity.
2. Load Evidence in Tool
    - Open Autopsy or FTK Imager.
    - Add the USB image file (E01, RAW or DD format) as evidence.
3. Scan for Deleted Files
    - Allow the tool to process the file system.
    - Check the Unallocated Space and Deleted Files sections.
4. Recover Files
    - Identify deleted files, such as .docx, .pdf and .jpg.
    - Export them to an evidence folder.
5. Verify Integrity
    - Generate MD5 or SHA-1 hash values for the recovered files.
    - Compare them with the original files, if available, to check integrity.
6. Document Findings
    - Prepare a forensic report containing:
        - File names and types

- File size and location
- Recovery status
- Hash values

STEP-BY-STEP PROCEDURE (Using Autopsy)

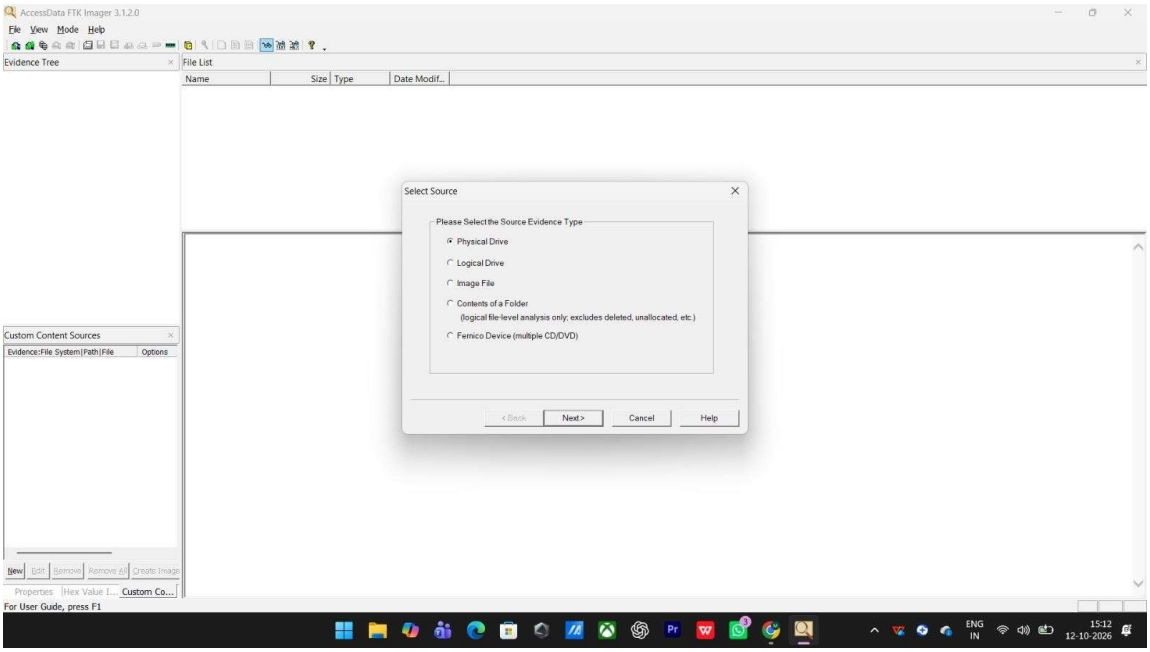
1. Launch Autopsy and create a new case.
2. Enter the case name as USB Deleted File Recovery.
3. Add the USB image file as the data source.
4. Configure ingest modules and enable File Type Identification and Deleted Files.
5. Start processing and wait for the analysis to complete.
6. Navigate to Views → Deleted Files.
7. Select the required files and use the available extraction or recovery option to save them in a secure evidence folder.
8. Use Autopsy's reporting feature to generate an HTML or PDF report.

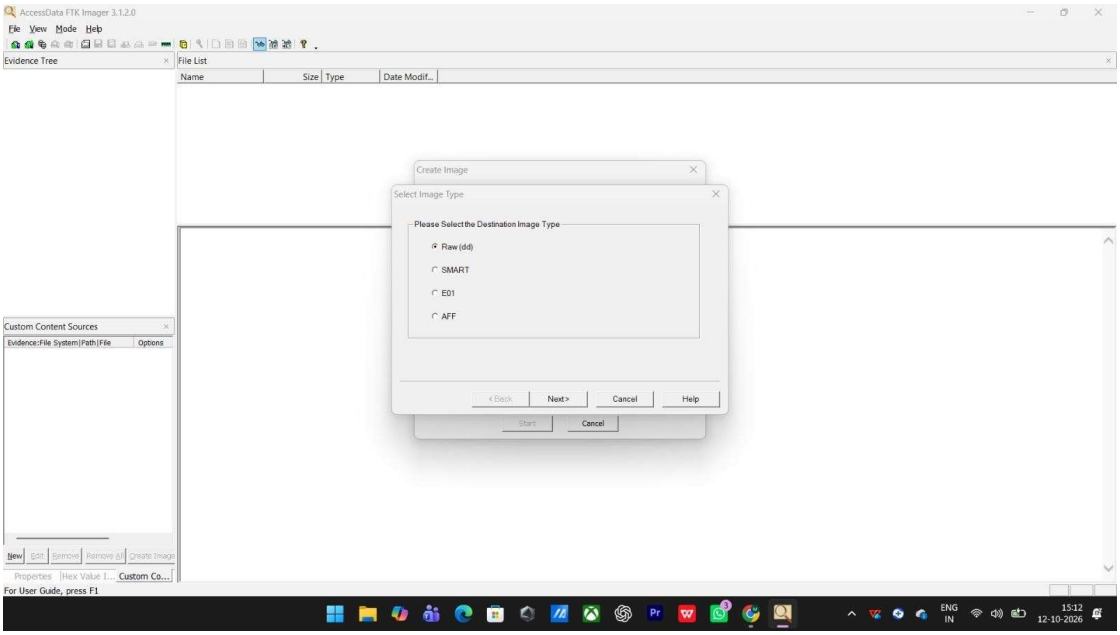
OBSERVATION

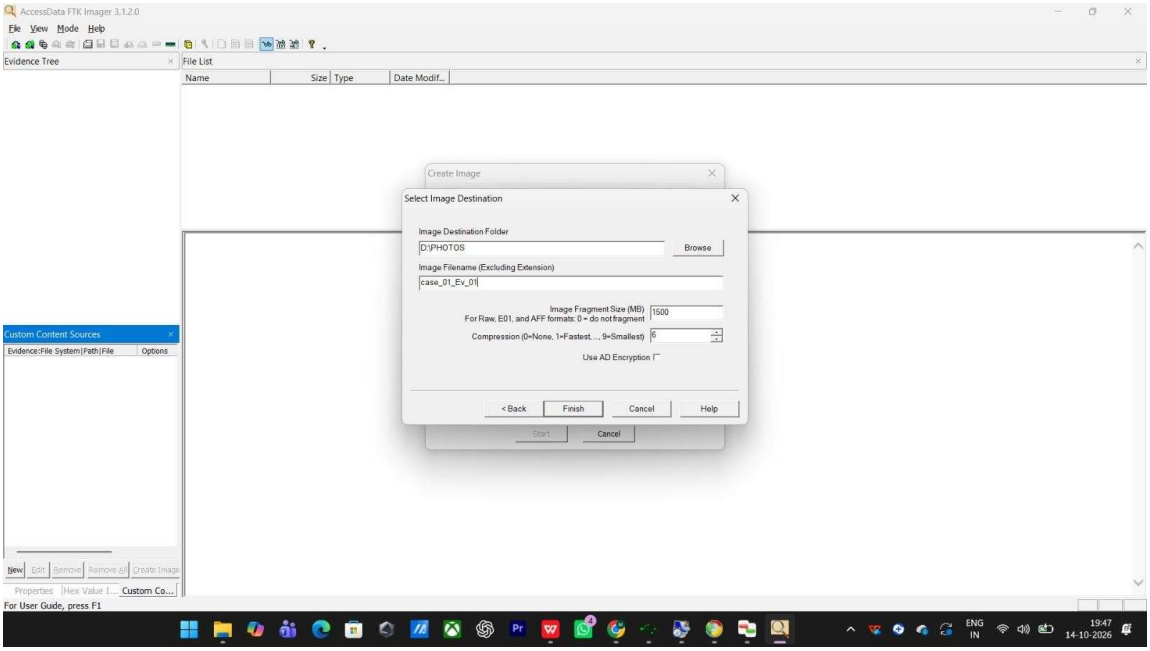
The USB image was loaded into the forensic tool and scanned for deleted files. Deleted files, if available, were identified and examined. The recoverable files were exported to an evidence folder, and their details were recorded for forensic analysis.

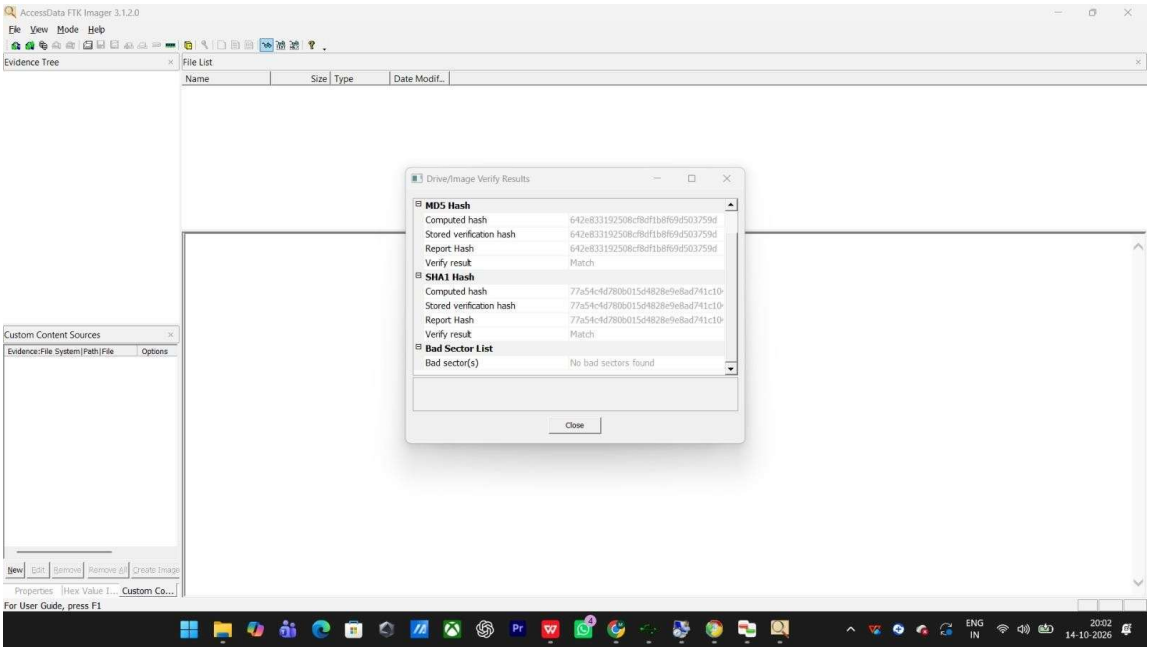
S.No.	Evidence Examined	Observation
1	USB image	USB forensic image was loaded into Autopsy.
2	Deleted files	Deleted files were examined in the Deleted Files section.
3	File types	File types such as DOCX, PDF and JPG were checked, where available.
4	Recovered files	Identified recoverable files were exported to an evidence folder, if available.
5	Hash values	MD5 or SHA-1 hash values were generated for integrity verification, if performed.
6	Forensic report	Findings were documented in a forensic report.

OBSERVATION









RESULT

Deleted files were searched for and recovered using forensic recovery tools, and the recovered data was examined successfully.

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EXPERIMENT 4

FIND LAST CONNECTED USB ON YOUR SYSTEM (USB FORENSICS)

AIM:

To identify the last connected USB device on the system using USBDeview and record its details for forensic investigation.

PROCEDURE:

STEP 1: DOWNLOAD USBDeview

1. Open a web browser.
2. Visit the NirSoft website.
3. Download the USBDeview ZIP file.

STEP 2: EXTRACT THE FILE

1. Locate the downloaded ZIP file.
2. Right-click and select Extract All
3. Open the extracted folder.
4. Find USBDeview.exe

STEP 3: Run USBDeview

1. Right-click on USBDeview.exe
2. Select Run as administrator
3. The USBDeview main window will open.

STEP 4: View USB Devices

1. Open USBDeview to view connected and previously used USB devices.
2. Check the device name, description, serial number, Vendor ID, Product ID, and last plug/unplug date.

STEP 5: Identify Last Connected USB

1. Check the Last Plug/Unplug Date column.
2. Compare the dates and identify the latest connection.
3. Record the device details in the observation table.

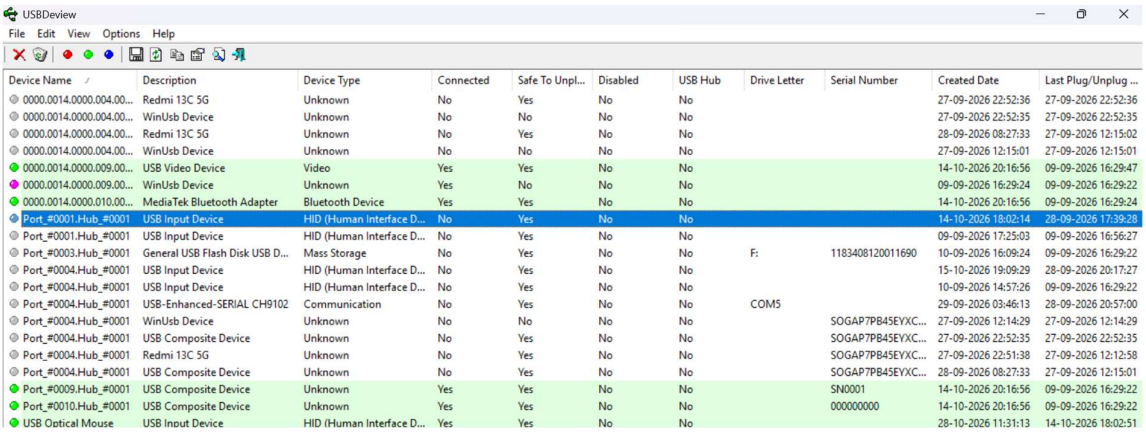
STEP 6: Record the Observation

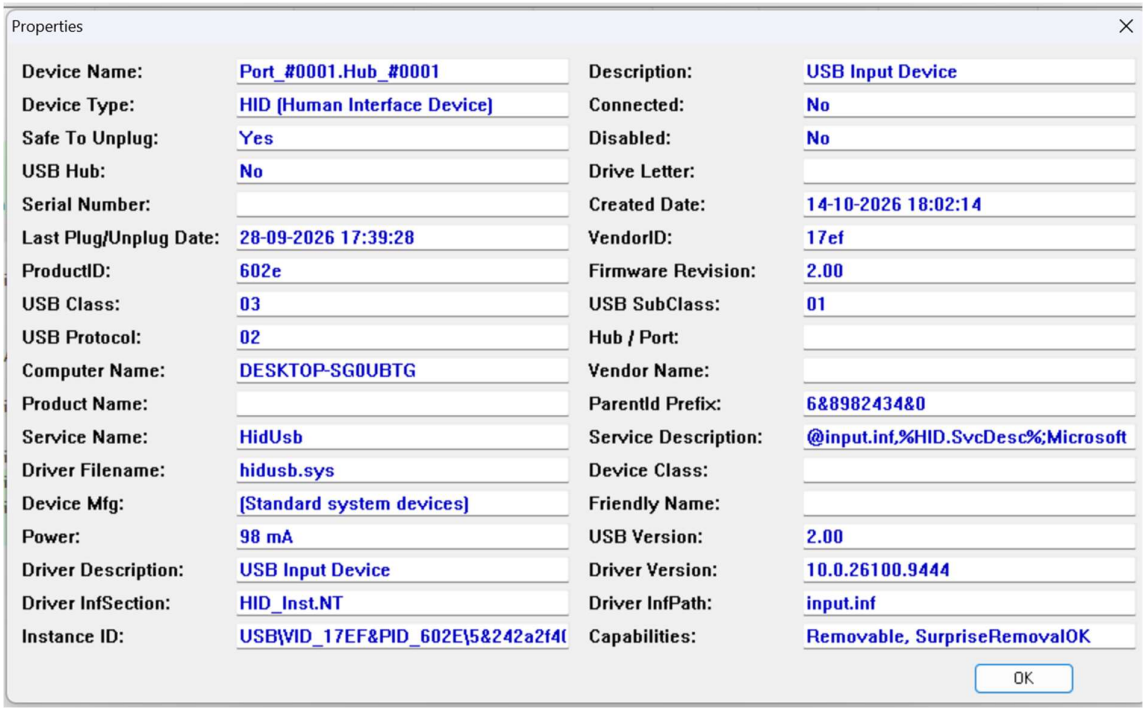
Enter the information displayed by USBDeview into the observation table in your record.

OBSERVATION:

The USBDeview tool was opened, and the USB devices previously connected to the system were displayed. The device details were examined, and the last connection date and time were recorded.

S.No.	DETAILS	OBSERVATION
1.	Device Name	USB Input Device
2.	Description	USB Input Device
3.	Device Type	HID
4.	Serial Number	C4S3000106912108117
5.	Last Plug/Unplug Date	28-09-2026 17:39:28
6.	Vendor ID	17ef
7.	Product ID	





RESULT:

Thus, the USBDeview tool was used to identify USB devices connected to the system, and the details of the last connected USB device were recorded successfully.

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EXPERIMENT 5

VIEW LAST ACTIVITY OF YOUR PC

AIM:

To identify and analyze the last activities performed on a computer using LastActivityView.

PROCEDURE:

STEP 1: Download LastActivityView

1. Open a web browser. Visit NirSoft website.
2. Download the LastActivityView ZIP file.

STEP 2: Extract the file

STEP 3: Run LastActivityView

1. Open LastActivityView.exe
2. Wait for the tool to collect available activity information.
3. The main window displays a list of recorded system and user activities.

STEP 4: View Recent Activities

1. Observe the list of activities displayed by the tool.
2. Check the activity name, date, time, and other available details.
3. Examine the events to understand the recent activity history of the computer.

STEP 5: Analyze Activity Details

1. Select an activity from the list. Check its record.
2. Compare the timestamps of different activities.
3. Identify relevant recent system events and user activities.

STEP 6: Record the Observation